Muhammad Sayyedul Awwab

Backend Software Engineer
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Summary

Backend software engineer with 4+ years in .NET Core. I build backend systems — the APIs, the data models, and the business rules underneath them — and I am working toward software architecture and technical leadership.

Professional Experience

- **Senior Programmer**
  Epyllion Group
  Dhaka, Bangladesh
  July 2025 – Present
  - Building an HRMS-style enterprise platform from the ground up in .NET Core with Clean Architecture, CQRS, DDD and the Result pattern, using Dapper over SQL Server, over operational data covering 25000+ employees.
  - Designed and implemented a scalable, workflow-driven Approval Engine supporting multi-step, parallel, and dependency-based approvals with dynamic approver resolution and timeout handling.
  - Designed and implemented a role-based access control (RBAC) system for navigation menus, pages, and UI elements, along with a user-level data permission framework enabling fine-grained, multi-dimensional access control across organizational hierarchies.
  - Architected a multi-channel notification system (Email, in-app via SignalR) on the Outbox pattern for at-least-once delivery, with Strategy-based payload builders and Scriban templates per workflow.
  - Identified and resolved inefficiencies in Azure DevOps CI/CD pipelines for a .NET Web API, reducing build pipeline time from **8m 18s to 4m 39s** and release time from **2m 12s to 51s**, cutting end-to-end merge and deployment time from **10m 30s to 5m 30s**.
  - Migrated data from the legacy database to the new schema using stored procedures.
  - Set up logs, metrics and traces with OpenTelemetry, Grafana, Loki, Tempo and Prometheus.
  - Reviewed teammates' code across the backend modules.

- **Software Engineer**
  Echologyx Ltd. UK
  Dhaka, Bangladesh
  Oct 2024 – July 2025
  - Worked on a UK e-commerce platform of 30+ interconnected .NET Core microservices, using CQRS and Clean Architecture.
  - Implemented inter-service communication with NServiceBus and wrote unit tests with xUnit.
  - Contributed to Azure DevOps CI/CD pipelines for automated testing and deployment, and reviewed teammates' code.

- **Junior Software Engineer**
  Bdjobs.com Ltd.
  Dhaka, Bangladesh
  Sep 2022 – Sep 2024
  - Resolved 200+ critical production issues in the employer system.
  - Owned the employer site's payment flows: job posting services, CV purchases, and SMS purchases.
  - Automated invoice generation for online payments of job posting services, reducing manual processing for the accounts team.
  - Led the migration of the Job Posting System from legacy technology to .NET Core Web API, decomposing legacy code into modular units so the team could develop in parallel.

- Migrated the Recruitment Center APIs to .NET Core using Clean Architecture, DDD and the Repository pattern.
- Built ETL background worker services with Hangfire to migrate and sync data from SQL Server to MongoDB.
- Collaborated with frontend and QA teams throughout migration, delivering features with 99% customer satisfaction.

PROJECTS

- **E-commerce Microservice -** *GitHub*
  - Designed a distributed e-commerce platform (Catalog, Identity, Ordering, Cart and a simulated Payment processor) behind a YARP API Gateway, using .NET Core, Clean Architecture, CQRS and DDD.
  - Order placement runs as an event-driven flow across Catalog, Payment and Ordering over RabbitMQ and MassTransit, with a transactional outbox so events are not lost.
  - JWT authentication with refresh tokens, Redis caching and SQL Server persistence; containerized with Docker, with observability through OpenTelemetry and the Grafana stack.
  - Documented the design decisions in four published articles.
- **Real-time Chat Service -** *Backend Service* | *Frontend Client*
  - Built the backend for one-to-one and group messaging over WebSockets with SignalR, covering auth, room management and conversations.
  - Applied Clean Architecture and CQRS, persisting users and rooms in SQL Server, conversations and messages in MongoDB, and SignalR connection state and recent messages in Redis.
  - A small React client was built only to exercise the API.

SKILLS

- **Backend & application architecture:** C#, .NET Core, REST APIs, Clean Architecture, Vertical Slice Architecture, CQRS, Repository pattern, Result pattern, Domain-Driven Design, Modular Monolith, Unit and integration testing (xUnit), FluentValidation, Serilog, Observability, OpenTelemetry, Grafana, Loki, Prometheus, Tempo, SignalR / WebSockets
- **Data:** Relational data modelling, SQL Server, PostgreSQL, EF Core, Dapper, MongoDB, Redis
- **Distributed systems & integration:** Microservices, Domain Events and Integration Events, RabbitMQ, MassTransit, NServiceBus, API gateway (Ocelot/YARP)
- **Delivery & operations:** Azure DevOps, CI/CD pipelines, IIS, Docker, JIRA, Git

WRITING

Articles on .NET Core, software architecture and databases at sayyedulawwab.com/blog and Medium.

EDUCATION

- **BSc in Information and Communications Engineering**
  Bangladesh University of Professionals (BUP), Dhaka | 2022 | CGPA: 3.68/4.00
- **Higher Secondary Certificate (HSC)**
  Saint Joseph Higher Secondary School, Dhaka | 2017 | GPA: 5.00/5.00